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CLINICAL REVIEW

Memory consolidation in sleep disorders

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SUMMARY

In recent years sleep-related memory consolidation has become a central topic in the sleep research field. Several studies have shown that in healthy individuals sleep promotes memory consolidation. Notwithstanding this, the consequences of sleep disorders on offline memory consolidation remain poorly investigated. Research studies indicate that patients with insomnia, obstructive sleep apnea, and narcolepsy often exhibit sleep-related impairment in the consolidation of declarative and procedural information. On the other hand, patients with parasomnias, such as sleep-walking, night terrors and rapid eye movement (REM) behavior disorder, do not present any memory impairment. These studies suggest that only sleep disorders characterized by increased post-learning arousal and disrupted sleep architecture seem to be associated with offline memory consolidation issues. Such impairments, arising already in childhood, may potentially affect the development and maintenance of an individual's cognitive abilities, reducing their quality of life and increasing the risk of accidents. However, promising findings suggest that successfully treating sleep symptoms can result in the restoration of memory functions and marked reduction of direct and indirect societal costs of sleep disorders.

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Introduction

Memory consolidation, the process by which labile information becomes stronger, more efficient, and more resistant to interference [1], is purported to occur during offline periods after encoding. As already observed in a pioneering study in 1924 by Jenkins and Dallenbach [2], this process seems to be facilitated when learning is followed by a sleep period. However, only in recent years has extensive research indicated that an intervening period of sleep after learning promotes the offline consolidation of declarative, perceptual, emotional, and procedural information [3]. Consequently, if sleep affects memory consolidation, it is reasonable to hypothesize that disturbed and/or altered sleep, as experienced by individuals with sleep disorders, may impair the ability to optimally consolidate information. Further, it has been estimated that about 45 million individuals in Europe [4], and between 50 and 70 million in the US suffer from sleep disorders [5], but very few research studies have investigated the consequences of sleep disorders on offline memory consolidation abilities. Yet, understanding the cognitive consequences of sleep disorders holds far-reaching implications for both future neuroscientific investigations as well as for clinical purposes. Nonetheless, only one review by Cipolli and colleagues [6] has been published on this topic. In that review, the authors supported the idea of an impaired memory consolidation process in patients with sleep disorders. The current review aims to present and integrate the newly available findings regarding sleep disorders and memory consolidation with the studies described in Cipolli and colleagues' work [6], providing a guide for future investigations trying to understand how memory impairments might be reduced via treatments targeting the sleep symptomatology, and thus, facilitating the translation from basic research to clinical practice.

The present review is organized as follows. First, I will briefly describe the main theoretical models regarding sleep and memory consolidation. Next, I will review all the available studies on sleep disorder and memory consolidation presented by type of disorder (i.e., insomnia, obstructive sleep apnea, narcolepsy, parasomnias) and the chief results will be outlined. Subsequently, these results

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Abbreviations

AHI	apnea-hypopnea index
ASRT	alternating serial reaction time task
CPAP	continuous positive airway pressure
FTT	finger tapping task
NREM	non-rapid eye movement
NC	narcolepsy with cataplexy
MA	motor adaptation
MSL	motor sequence learning
MTT	mirror tracing task
OSA	obstructive sleep apnea
PSG	polysomnography
RBD	REM behavior disorder
ROCFT	Rey–Osterrieth complex figure test
REM	rapid eye movement
SOREM	sleep onset rapid eye movement
SMR	sensorimotor rhythm
SWA	slow wave activity
SWS	slow wave sleep
TDT	texture discrimination task
WAIS	Wechsler adult intelligence scale
WPA	word-pair association task

will be discussed within a comprehensive perspective on how memory consolidation is affected by sleep disturbances. In the final section, clinical implications will be advanced and discussed.

Sleep and memory consolidation

Offline memory consolidation is typically assessed comparing performance in a memory task before and after a period of time that includes sleep and/or wakefulness. Several theoretical models have been proposed to explain the role of sleep in the consolidation of information. Beyond the dual process hypothesis [7] and the opportunistic consolidation theory [8], the models that so far have attracted the most interest in sleep research are the active system consolidation model [1] and the synaptic homeostasis hypothesis [9].

The active system consolidation model [1], based on the two-stage model of memory trace formation [10], proposes that during wakefulness, stimuli are initially encoded in parallel in two different memory systems that store information at different rates: in the hippocampus at a faster rate and in cortical networks at a slower rate. During the subsequent sleep period, specifically during non-rapid eye movement sleep (NREM, composed by stage N1, N2 and N3, the latter also known as slow-wave sleep, SWS), the pieces of information learned during wake are re-activated in both the hippocampus and in the cortex. These reactivations, through a constant dialog between the hippocampus and the neocortex, mediate the redistribution of information in cortical areas. These cortical connections are then strengthened and stabilized via synaptic consolidation, and eventually integrated with pre-existing knowledge, making the learned event less susceptible to interference [3,11]. In addition, consolidated information becomes independent from the hippocampus, which returns to an optimized state for the encoding of new material [12–14]. This model provides an elegant framework for the consolidation of hippocampus-dependent information such as declarative and explicit motor memories [3].

According to the synaptic homeostasis hypothesis [9], the encoding of information during wakefulness leads to increased strength of connections (i.e., synaptic weights). During the subsequent sleep, when external inputs are reduced, slow oscillations renormalize these synapses inducing synaptic depression. This leads to a weakening of unimportant and less integrated information, making the important (signal) relative to the spurious information (noise) more salient. This process also restores the capacity of synapses to acquire new information. Since memory formation and stabilization requires some forms of synaptic plasticity, this model offers a parsimonious explanation for the consolidation of both hippocampal (i.e., declarative memories) and non-hippocampal information (e.g., perceptual learning) [9].

Recently, it has been proposed that these models may not be mutually exclusive, in the sense that they can describe complementary processes that improve memory consolidation during sleep [15–17]. Indeed, it is possible that during NREM sleep, salient information associated with experience acquired in wake is reactivated and integrated in cortical networks [15]. This process seems to be driven by the temporal coupling of thalamo-cortical sleep oscillations, such as slow oscillations, sleep spindles and sharp-wave ripples [18–21]. In the subsequent SWS, memory reactivations can be alternated with the synaptic downscaling process in order to reduce the strength of neural networks [16], resulting in an overall synaptic reduction [9]. However, previously integrated information remains strong, thus increasing signal-to-noise ratio and reinforcing reactivated memory traces [9].

In light of these theoretical models, it is reasonable to expect that when sleep patterns are disturbed and/or altered, such as is the case of sleep disorders, the ability to optimally consolidate information may be impaired. However, as it will be described in the following paragraphs (see also Table 1), the available findings on memory consolidation and sleep disorders are quite heterogeneous.

Memory consolidation in insomnia

Insomnia is a very common sleep disorder, with prevalence rates ranging 6–10% in the general adult population [22]. Insomnia-dependent cognitive impairments, included among the diagnostic criteria of both ICD-3 [23] and DSM-5 [24], have been well described (for a review [25]). Nevertheless, memory consolidation in insomnia remains poorly investigated.

In 2006, Backhaus and colleagues hypothesized that individuals with primary insomnia would show impairment in the consolidation of both procedural (motor adaptation abilities, MA) and declarative memories. To test this idea, they recruited 16 primary insomniacs (M_{age} = 41.6 y) and 13 controls (M_{age} = 40.1 y) and had them participate in two laboratory polysomnographic recordings (PSG), an adaptation and an experimental night, respectively [26]. The evening prior to the experimental night (learning phase), participants performed a word-pair association task (WPA) and a mirror-tracing task (MTT; Figs. S1–2). In the subsequent morning, participants performed a retrieval phase per task. Although the two groups displayed a similar performance during the learning phase, the number of word-pairs recalled at retrieval remained stable for insomniacs (but increased for controls), whereas procedural memories remained stable in both groups. Nevertheless, it should be noted that the WPA paradigm used in this study included the presentation of the correct word-pair after the last retrieval session in the evening, before sleep. Consequently, this did not allow them to parse whether controls improved their performance from the level reached at the end of the learning session, or if they just showed a slower decay than insomniacs.

Table 1
Demographics and main results of the studies investigating memory consolidation in sleep disorder by type of disorder.

Disorder	Study	Sample (*Mage*±SD)	Declarative task*	Correlations declarative task	Procedural task*	Correlations procedural task	Daytime control
Insomnia	Backhaus et al., 2006 [26]	16 PI (41.6 ± 1.2) 13 HS (23.1 ± 2.4)	WPA (PI < HS)	REM_{PI} (+) SWS_{HS} (+)	MTT (PI ≈ HS)	ns	no
	Nissen et al., 2006 [27]	7 PI (44.9 ± 4.1) 7 HS (44.3 ± 5.3)	–	–	MTT (PI < HS)	$REMD_{ALL}$ (+) TST_{ALL} (−)	no
	Nissen et al., 2011 [28]	33 PI (range 37–58) 53 HS (range 40–58)	VVLT (PI < HS)	ns	MTT (PI < HS)	$REMD_{HS}$ (+)	yes
	Göder et al., 2011 [43]	15 PSI, 3 IDI, 13 PXI, 6 HYS, 5 DD AGE_{ALL} (*Mage* 40.3, range 16–67)	WPA‡ ROCFT‡	ns TST_{ALL} (+), $NREM_{ALL}$ (+), SE_{ALL} (+), $NREM\text{-}REMcycles_{ALL}$ (+)	MTT‡	ns	no
	Griessenberger et al., 2013 [42]	27 PI (36.7 ± 10.5) 21 HS (33.5 ± 9.8)	WPA (PI < HS)	REM_{HS} (+), $WASO_{HS}$ (−) AWN_{PI} (−), $SWSspindlesACT_{PI}$ (+)	FTT (PI ≈ HS)	REM_{HS} (−) $SWSspindlesACT_{HS}$ (+)	no
	Schabus et al., 2013 [100]	24 PI (34.8 ± 10.6)	WPA‡	ns	–	–	no
	Cellini et al., 2014 [32]	13 PI (23.3 ± 2.6) 13 HS (24.3 ± 1.6)	–	–	FTT (PI < HS)	ns	no
OSA	Kloepfer et al., 2009 [46]	15 OSA (46.4 ± 5.9) 20 HS (47.4 ± 5.6)	VVLT (PI < HS)	$REMcycles_{OSA}$ (+)	MTT (OSA < HS)	ns	no
	Mendes Medeiros et al., 2012 [47]	20 OSA (57.9 ± 5.8) 10 HS (56.4 ± 6.3)	WPA (PI < HS) Maze test (PI ≈ HS) Emotional memory (PI ≈ HS)	ns SWS_{OSA} (−) ns	–	–	no
	Djonlagic et al., 2012 [50]	16 OSA (31.9 ± 1.7) 15 HS (29.0 ± 1.7)	–	–	FTT (OSA < HS)	ArI_{OSA} (−) AHI_{OSA} (−)	no
	Djonlagic et al., 2014 [51]	20 OSA (41.1 ± 1.0) 20 HS (35.3 ± 2.6)	–	–	FTT (OSA < HS)	ArI_{OSA} (−) AHI_{OSA} (−)	no
	Landry et al., 2014 [52]	12 OSA (53.0 ± 1.8) 12 HS (52.8 ± 1.9)	–	–	FTT (OSA < HS)	$WASO_{HS}$ (−)	no
	Csabi et al., 2014 [49]	17 OSA (52.4 ± 9.7) 17 HS (54.2 ± 7.3)	–	–	ASRT (OSA < HS)*	ns	no
	Djonlagic et al., 2015 [53]	12 OSA (43.3 ± 13.0) 14 CPAP-1 (44.4 ± 11.1) 12 HS (37.3 ± 10.5)	–	–	FTT (OSA/CPAP-1<HS; OSA ≈ CPAP-1)	ns	no
	Landry et al., 2016 [54]	15 OSA (47.2 ± 1.7) 13 CPAP-1 (47.7 ± 1.9) 17 CPAP-c (48.5 ± 2.0) 14 HS (47.0 ± 2.7)	–	–	FTT (OSA/CPAP-1< CPAP-c-HS; CPAP-c ≈ HS)	$ArI_{CPAP\text{-}1}$ (−) $ArI_{CPAP\text{-}c}$ (−) $AHI_{CPAP\text{-}1}$ (−)	no
SDB	Kheirandish-Gozal et al., 2009 [56]	54 OSA (10.5 ± 0.5) 17 HS (12.1 ± 0.9)	PMT (OSA < HS)	ns	–	–	no
	Csabi et al., 2016 [57]	16 SDB (8.6 ± 2.3) 16 HS (8.8 ± 1.4)	Story recall (SBD ≈ HS)	ns	ASRT (SBD < HS)	ns	no
NC	Cipolli et al., 2009 [62]	22 NC (30.2 ± 6.6) 22 HS (29.7 ± 6.1)	–	–	TDT (NC < HS)		no
	Mazzetti et al., 2012 [38]	14 NC (31.4 ± 8.4) 14 HS (30.9 ± 7.1)	–	–	FTT (NC < HS)		no

(*continued on next page*)

Table 1 (*continued*)

Disorder	Study	Sample (*Mage*±SD)	Declarative task*	Correlations declarative task	Procedural task*	Correlations procedural task	Daytime control
Parasomnia	Oudiette et al., 2011 [71]	19 SW (34.4 ± 15.4) 20 RBD (66.5 ± 6.5) 18 HS (57.9 ± 5.3)	–	–	SRTT (SW ≈ RBD ≈ HS)	ns	no
	Uguccioni et al., 2013 [72]	18 RBD (67.0 ± 7.6) 10 HS (64.2 ± 9.3)	FCRST (RBD ≈ HS) Story Recall (RBD ≈ HS)	ns	–	–	yes
	Uguccioni et al., 2015 [79]	19 SW (30 [26,34])† 19 HS (32 [27,36])†	FCRST (RBD ≈ HS) Story Recall (SW ≈ HS)	ns	–	–	yes

Notes. AWN: numbers of awakenings; AHI: apnea-hypopnea index; ALL: all participants; ArI: arousal Index; CPAP-1: first night CPAP users; CPAP-C: compliant CPAP users; DD: dysthymic disorder; FCSRT: free and cued selective reminding test; FTT: finger tapping task; HS: healthy sleeper; HYP: hypersomnia; IDI: idiopathic insomnia; Mage: mean age in years; MTT: mirror tracing task; NC: narcolepsy-cataplexy; NREM: non rapid eye movement; ns: non-significant result(s); OSA: obstructive sleep apnea; PMT: pictorial memory task; PI: primary insomnia; PSI: psychophysiological insomnia; PXI: paradoxical insomnia; RBD: REM sleep behavior disorder; REM: rapid eyes movement; REMD: REM density; ROCFT: Rey–Osterrieth complex figure test; SD: standard deviation; SDB: sleep-disordered breathing; SE: sleep efficiency; SRTT: serial reaction time task; SW: sleepwalking; SWS: slow wave sleep; TST: total sleep time; VVLT: visual and verbal memory task; WASO: wake after sleep onset; WPA: word-pair association task; (+): positive correlation, (−): negative correlation; ‡: no groups comparisons; †: median age [lower quartile; upper quartile]; *: general performance differences have been reported for descriptive purposes and these differences may not reflect specific significant effects.

Also, this study (as several others included in this review, see Table 1) did not include a daytime control condition (i.e., a period of waking consolidation). The exclusion of this control condition makes it difficult to attribute the observed differences to sleep after learning. Consolidation differences might also occur during periods of daytime consolidation, indicating that the memory impairment is not due to sleep *per se*, but may depend on other characteristics of patients.

Nissen and colleagues [27], testing seven individuals with primary insomnia and seven controls in a procedural task (i.e., the MTT), observed a reduced gain in overnight consolidation of procedural skills in insomniacs (~20%) as compared to controls (~43%). These authors also extended their research to a group of 33 primary insomniacs (M_{age} = ~46 y) and 53 controls (*M*age = ~47 y) tested with the MTT plus a visual and verbal memory task in both a sleep (12 h of nighttime consolidation) and a wake condition (12 h of daytime consolidation, condition which was not present in their previous study) [28]. Although controls showed a significant improvement in retrieving both procedural and declarative information during a 12-hour sleep period as compared to wake, the two groups did not show any differences in the daytime consolidation. Insomniacs did not exhibit any overnight benefit.

Impaired memory consolidation was also observed for a different type of procedural learning: explicit motor sequence learning (MSL). Differently from MA skills, which require the modification of previously acquired movements in response to environmental changes, MSL involves the integration of sequences of movements into a coherent unit [29,30]. A widely-used explicit MSL task is the finger tapping task (FTT; Fig. 1). Indeed, post-training performance in this task usually stabilizes after a period containing sleep [31–33].

Using the FTT Cellini and colleagues [34] tested 13 young primary insomniacs (M_{age} = 23.31 y) and 13 controls (M_{age} = 24.31 y) before and after a night of polysomnographic recording in the lab. During the pre-sleep evening training, although insomniacs performed worse than controls, their learning curve (i.e., the trajectory of improvement from the first to the last three blocks of the evening training) was equivalent, indicating a similar online motor learning between groups. In the morning, controls' performance was enhanced by ~22% compared to the performance level reached at the end of the evening training, in line with other reports of an overnight effect on motor skills consolidation [33,35–41]. Insomniacs failed to exhibit any offline enhancement (only ~8%). Due to the lack of a daytime condition in this study, the authors could not distinguish whether the observed absence of motor skills consolidation in insomnia was sleep-related or just time-related.

Explicit MSL and declarative memory were also assessed in a study aimed to test susceptibility to memory interference [42] in 27 primary insomniacs (M_{age} = 36.67 y) and 21 controls (M_{age} = 33.53 y). In the evening of each of two experimental nights, participants were trained either in a FTT or in a WPA task. After a retrieval morning session, participants performed an interference task (i.e., either a different FTT sequence or new word-pairs). This was followed by a second retrieval session of the originally learned task in order to assess the susceptibility to interference of the consolidated material. No daytime condition was present in this study. Insomniacs (vs. controls) showed a reduced overnight benefit, and no significant interference effect for procedural memories. However, despite similar consolidation levels, after the interference manipulation insomniacs remembered less paired words compared to controls. Interestingly, the interference effect was observed also in a sub-group of participants tested 5–8 d following the second experimental night. These results indicate that declarative memories are more susceptible to interference in insomniacs.

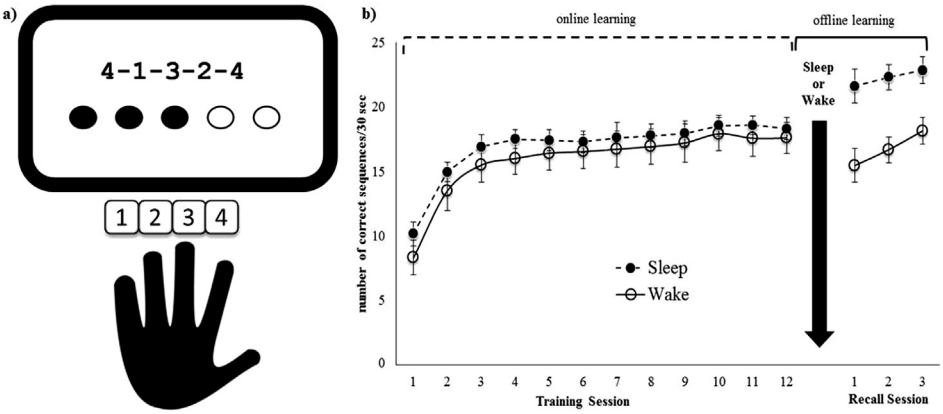


Fig. 1. a) The finger tapping task (FTT) is a motor sequence learning task in which participants are asked to tap with their non-dominant hand a 5-digits sequence (for example 4-1-3-2-4 or 2-4-1-3-2), as rapidly and accurately as possible, using the numeric key-buttons of a computer keyboard or a response box. To reduce working memory load, both the numeric sequence and five white circles are displayed at the center of the screen throughout the task. Every time the participants press a key, one circle turns into a black dot. b) Example of a FTT paradigm. During a training session, participants perform 12 blocks, each consisting in 30s of tapping followed by 30s of rest. Then, after a period containing sleep or just wakefulness, participants perform a recall session, which can be composed by 3–12 blocks. The change in performance throughout the training session is considered a measure of online (or practice-dependent) learning, whereas the change in performance from the last three blocks of learning to the blocks in the recall session represents an index of offline memory consolidation. In this figure the number of correct sequences per 30s is reported in two hypothetical groups, sleep and wake.

In a correlational study, a sample of adults with psychophysiological, idiopathic and paradoxical insomnia, mild hypersomnia, and dysthymic disorder, performed both a mirror tracing and two declarative tasks - WPA and the Rey–Osterrieth complex figure test (ROCFT) - before going to sleep and the following morning [43]. Participants did not show any improvements in either WPA (probably due to a ceiling effect caused by the low amount of pairs used) or in the ROCFT, whereas they improved in the MTT (~40%). Morning performance in the ROCFT was positively associated with total sleep time, sleep efficiency, NREM sleep time and the number of sleep cycles. The same results were observed when the insomnia subgroup was analyzed alone. This study showed an interesting relationship between sleep quality and post-sleep performance in the visual memory domain in patients with sleep disorder. However, the lack of a control group did not allow us to understand whether these patients experienced any degree of memory impairments.

Overall, these studies indicate that insomniacs experience difficulties in consolidating declarative memories [26,28,42], whereas for procedural memory the findings are less clear. In Backhaus et al. [26] no difference was observed for the overnight consolidation of MA abilities between insomniacs and controls, whilst in Nissen et al. [28] insomniacs showed an impaired offline consolidation. These contradicting results may depend on several factors, such as heterogeneity of the sample for age, comorbidity and educational level, as well as for the task procedure. Indeed it is worth noting that the overnight change in the mirror tracing time varied across all these studies (Backhaus: ~33%; Nissen: ~21%; Goder ~40%). Similarly, the two studies using the FTT showed inconsistent results. Cellini et al. [34] reported impaired offline performance in young adults with insomnia, whereas Griessemberger et al. [42] reported similar overnight consolidation in insomniacs and controls. This inconsistency could be due to the different way of computing FTT performance in the studies (i.e., using the entire five-element sequence vs. using only a three-element sequence out of five), as well as to the differences in sample age. For the latter point, it is worth noting that performance in the FTT is reduced in healthy participants over 30 y old relative to younger individuals [37].

Memory consolidation in obstructive sleep apnea

Obstructive sleep apnea (OSA) is a common sleep disorder characterized by intermittent hypoxia that impacts sleep quality inducing frequent arousals and sleep fragmentation. OSA has been associated with impairments in cognitive functioning [44], although objective measures do not always support subjective complaints [45].

In 2009, Kloepfer and colleagues [46] investigated declarative and procedural memory consolidation in 15 middle-aged patients with moderate OSA (M_{age} = 46.4 y; apnea-hypopnea index [AHI] of 15–25 events/h) and 20 controls (M_{age} = 47.4 y). The procedure was similar to another study by the same group focusing on insomnia and described above [28]. Participants spent one night in the lab undergoing a standard PSG investigation. In the evening they performed the MTT, a visual and verbal memory task, and a retest session occurred the subsequent morning. The study did not include a daytime condition. Although groups did not show any significant difference in the performance of both tasks during the evening session, offline consolidation of verbal and visual materials was nominally reduced in OSA patients as compared to controls. Also, OSA patients exhibited a reduced overnight improvement in the MTT relative to controls (30.1% vs 39.5%, respectively). However, by analyzing the evening and the morning MTT performance trial-by-trial (instead of using the averaged performance), authors showed that OSA patients had a slower online performance in both sessions, but the absolute overnight enhancement did not differ between groups.

Medeiros and colleagues [47] assessed verbal, visuo-spatial and emotional memory consolidation in 20 patients with moderate/severe OSA (M_{age} = 57.9 y; AHI > 15 events/h) and 10 controls (M_{age} = 56.4 y). Participants performed an encoding session in the evening and a retrieval session the following morning. The WPA and the maze test from the Wechsler adult intelligence scale (WAIS, [48]) were used to assess verbal and visuo-spatial memory, respectively, whereas an emotional picture recognition task was used to assess emotional memory. No daytime condition was present in this study. While controls improved their morning performance in the visuo-spatial task, performance of OSA patients worsened. No difference between groups was observed for the emotional memory task. In contrast to Kloepfer and colleagues [46], patients with OSA in this study did not show any impairment in verbal memory relative to controls, but only in visuo-spatial memory. This may be the consequence of the different tasks employed in the studies, with the WAIS subtests possibly not being sensitive enough to assess memory consolidation.

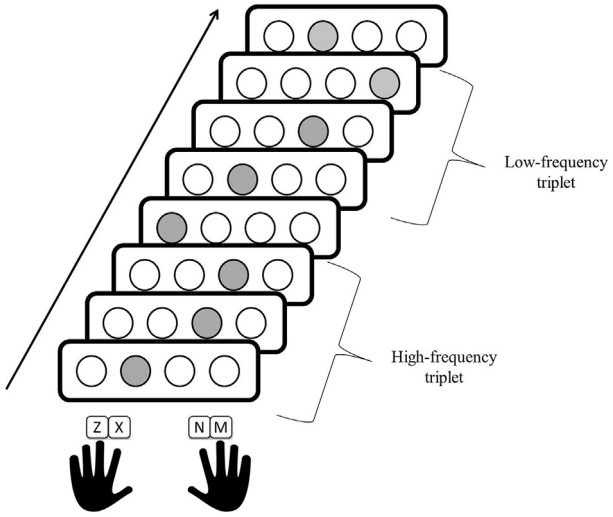


Fig. 2. Example of an alternating serial reaction time (ASRT) task sequence. In the ASRT a stimulus (e.g., a colored dot, a dog's head) appears in one of four circles presented on the screen. Participants have to respond as quickly and accurately as they can using the corresponding respond button (e.g., "Z", "X", "N", "M" buttons). Trials are composed of 8-stimuli sequences. Participants are not told that these sequences are composed of triplets of stimuli with high or low probability to occur. For example, when "2" and "3" occur in sequence, "3" is followed by another "3" on 4% of the times (high-frequency triplet), and by either "1", "2" or "4" in 0.8% of the times (low-frequency triplets). In Csabi's studies [49,57], of the possible 64 triplets, 16 were high-frequency triplets (occurring in the 62.5% of the trials) and 48 low-frequency triplets (occurring in the 37.5% of the trials). Implicit motor learning is measured as the change in responses' reaction times across blocks (online learning) and/or across sessions (offline learning), considering both the responses to all stimuli ("general learning") and only for the high-frequency relative to the low-frequency triplets (sequence-specific learning).

Another study evaluated the consolidation of procedural information in OSA patients using an implicit MSL task [49]. Unlike the FTT, in which participants tap an explicit and fixed motor sequence, in the alternating serial reaction time task (ASRT), the sequences vary trial-by-trial in a probabilistic fashion, with some sequences presented with a higher frequency than others (Fig. 2).

Seventeen first-time diagnosed OSA patients (M_{age} = 52.41 y) and 17 healthy controls (M_{age} = 52.24 y) performed a learning session of the ASRT in the evening, followed by a standard PSG sleep study, and a retrieval session the following morning. This study did not include a daytime condition. In the evening, online learning did not differ between OSA patients and controls for both general learning (overall performance) and sequence-specific learning (performance to the high-frequency sequence relative to the low-frequency sequence). However, change in offline general ability was lower in OSA patients relative to controls, but they were still able to successfully learn and consolidate a probabilistic motor sequence (i.e., high-frequency triplets).

A series of studies have focused only on explicit motor memory consolidation in OSA patients using the FTT. Sixteen patients with mild OSA (M_{age} = 31.9 y) and 15 controls (M_{age} = 29.0 y) were tested on their consolidation of explicit MSL [50]. OSA patients were first-diagnosed with an AHI>5 events/h and no prior exposure to continuous positive airway pressure (CPAP) treatment. Participants performed the FTT in the evening before undergoing a PSG sleep recording in the lab. The next morning participants were first re-tested in the sequence they were trained on the previous evening, then they were requested to learn a new motor sequence to control for circadian effects of learning. No wake consolidation condition was included. Despite the two groups similar online learning performance during the evening training, OSA patients showed a significantly reduced overnight consolidation improvement compared to controls (1.1% vs 14.7%, respectively). However, OSA patients did not show a different performance relative to the controls for the new sequence learned in the morning. Interestingly, the authors observed a negative relationship between overnight improvement and arousal index, indicating that the offline consolidation process was affected by the number of nocturnal arousals.

A subsequent study by Djonlagic and colleagues [51] confirmed that the offline consolidation of explicit motor skills is impaired in patients with OSA. Twenty first-diagnosed OSA patients (M_{age} = 35.3 y) and 20 controls showed a similar online learning in the FTT both at evening and the next morning, but the OSA patients did not benefit from the overnight consolidation period. Again, the arousal index was negatively associated with the offline performance change, suggesting that memory consolidation is affected by sleep fragmentation. Moreover, only in patients with OSA, the severity of the disorder (AHI values) and age significantly predicted the magnitude of the overnight performance change.

A similar result was also reported by Landry and coworkers [52]. No significant differences were observed for the evening online learning in the FTT between 12 first-diagnosed OSA patients (M_{age} = 53 y) and 12 controls (M_{age} = 52.8 y). Also, while the controls showed about 15% of overnight improvement in the number of sequences correctly tapped, the OSA patients did not benefit from the offline consolidation period (performance increase of about 2%). Again, no daytime consolidation condition was included in the study.

More recently, Djonlagic and colleagues [53] assessed the effect of first-time experience with CPAP on the consolidation of procedural motor memory. Using the same paradigm used in their previous studies [50,51], they tested 15 patients who underwent a whole night with CPAP for their first time (CPAP group; M_{age} = 44.4 y), 14 OSA patients without CPAP and with no prior exposure to CPAP (no-CPAP group; M_{age} = 43.3 y) and 15 healthy controls (M_{age} = 37.3 y). Although the CPAP group reported a better subjective sleep quality, and showed faster reaction time in a vigilance task than the no-CPAP group, the two OSA groups did not exhibit any overnight motor improvement compared to controls, regardless of the night spent with or without the CPAP.

These results were further corroborated by Landry and colleagues [54]. The authors assessed the effect of CPAP treatment on the overnight consolidation of motor skills in 17 compliant CPAP users (M_{age} = 48.5 y) relative to 24 patients with untreated OSA (M_{age} = 47.2 y), 13 patients with first-night CPAP (M_{age} = 47.7 y), and 14 controls (M_{age} = 47.0 y). In contrast to the previous studies, participants performed a first test session 10 min after an evening training session in the FTT and before undergoing a nocturnal PSG study. This design allowed the authors to test whether the possible change in performance was merely due to the overnight consolidation period, or to other factors, such as fatigue and time-on-task effects [31]. Indeed, for this specific MSL task, recent studies showed that when the testing session occurred 5–30 min after a training session, in the absence of any intervening sleep period, performance increases by about the same magnitude observed after a period of sleep [33,55]. The next morning participants were re-tested on the same task. At the end of the evening session the four groups showed neither significant differences in offline nor online learning. Also, when re-tested after 10 min, all groups showed a significant performance enhancement (13%–22%). However, the overnight performance change differed between groups. Considering only the performance change between the end of the evening training session and the morning test, OSA and first-night CPAP patients showed negative or no overnight improvement (−4.86 and +1.6%, respectively), whereas compliant CPAP patients and controls significantly benefitted from the overnight period (+8.9%

and +11.6%, respectively). However, when considering the performance change between the post-training and the morning retest sessions, the performance of all four groups significantly deteriorated, and no significant differences between groups were shown. Visual inspection of the data showed that compliant CPAP users and controls exhibited a reduced degree of deterioration over the sleep period. As suggested by the authors, the absence of a significant effect was probably due to a lack of statistical power. Correlational analysis showed a negative correlation between arousal index and performance change for the CPAP users (both first-night and compliant patients). This study showed that motor skills improved in OSA patients only 10 min after the initial training, indicating a preserved ability to acquire and consolidate short-term new motor information. However, during the subsequent overnight period non-treated OSA and first-night CPAP patients were not able to stabilize this information. It is possible that in both non-treated OSA patients and first-night CPAP users, cellular consolidation processes, such as early long-term potentiation and synaptic morphogenesis [8] - which occurs just after encoding new motor information - are preserved, whereas systems related to sleep consolidation, may be impaired. Therefore, although sleep may help to stabilize this fragile motor information process in controls and compliant CPAP users, untreated OSA patients and first-time CPAP users do not benefit from this long-term consolidation process. However, given the lack of a daytime control condition, the extent to which this learning was specifically related to sleep remains unclear. Nevertheless, this study shows for the first time that memory consolidation process may be restored to an optimal level when the sleep symptomatology is successfully treated.

Children with sleep-disordered breathing

Memory consolidation seems to be impaired also in children with sleep-disordered breathing. Kheirandish-Gozal and colleagues [56] assessed the encoding and consolidation of declarative stimuli (i.e., pictures of animals) in a large sample of children with OSA. Fifty-four children with OSA (M_{age} = 10.5 y) and 17 controls (M_{age} = 12.1 y) were asked to identify the animals depicted in 26 colorful pictures. Children were asked to recall those animals' pictures (divided in four trials) 10 min after the learning session (immediate recall) and the following morning upon awakening (delayed recall). Children with OSA showed a constantly lower performance, and a reduced performance improvement from the first to the fourth trials, compared to the controls. In addition, the next morning, while controls maintained the performance level reached in the evening, the performance of children with OSA was further deteriorated. After controlling for potential confounders, the authors observed that the arousal index was the best predictor of performance, with higher arousal levels associated with lower memory performance. Some aspects of this study should be taken into consideration. Firstly, although not statistically significant, children with OSA were substantially younger than controls (10.5 vs 12.1 y). The difference of 1.5 y in this critical developmental period, coupled with a markedly different number of subjects per group (54 patients, 17 controls), may have affected the comparison between the groups. Also, this study did not include a daytime control condition, which did not allow for a clear understanding of the role of sleep factors in the observed group differences.

More recently, Csabi and coworkers [57] investigated the overnight consolidation of both verbal memory and implicit motor sequence performance in 16 untreated children with sleep-disordered breathing (M_{age} = 8.56 y) compared to 16 controls (M_{age} = 8.75 y). In the evening, participants performed a modified version of the ASRT task and listened to a short story that they were asked to repeat (immediate recall). The next morning, after 12 h including an overnight sleep period, children were retested in the same tasks. Although no differences were observed between groups in the implicit motor sequence task, children with sleep-disordered breathing performed worse than controls in remembering the story both in the evening and morning sessions. Their performance, which nominally worsened in the morning compared to the previous evening, may reflect a morning floor effect. These findings indicate that difficulties in the encoding and consolidation of declarative information are related to sleep-related breathing issues even in children, whereas the ability to learn and consolidate an implicit motor sequence seems to be preserved. The latter results may not be surprising considering that the magnitude of offline consolidation of motor skills in children seems to be similar after either a period of wake or sleep [3]. However, the lack of a daytime condition in the current study did not allow the confirmation of these results also in children with OSA. In addition, since it has been proposed that higher encoding level is associated with greater sleep-dependent consolidation (see [58]), further studies including a daytime control condition are required to better understand if, and to what degree, the lack of sleep benefits observed in children with OSA are due to sleep *per se* or to a general reduced ability to acquire new declarative information.

Memory consolidation in narcolepsy

Narcolepsy is a less prevalent disorder than insomnia and OSA, affecting approximately 0.05% of the general population [59], but negatively influencing several aspects of their lives [60]. Narcolepsy is characterized by excessive daytime sleepiness, the possible presence of cataplexy and by distinct differences in sleep architecture relative to healthy individuals (i.e., sleep fragmentation, sleep onset in REM (SOREM), reduced SWS; [61]).

To date, two studies from the same group have investigated memory consolidation in narcolepsy with cataplexy (NC) patients. Cipolli and coauthors [62] investigated the ability of NC to consolidate perceptual abilities. Twenty-two first-time diagnosis NC patients and 22 aged-matched controls were recruited in this study. The texture discrimination task (TDT), a task shown to be highly sensitive to post-training sleep [63,64], was employed to assess visual discrimination skills. To control for possible circadian influences in encoding, half of the participants performed the TDT at 11:00 h, whereas the other half at 17:00 h of the first experimental day. Then, all the participants performed the task again the next day at 11:00 h. Seven days later, participants were asked to perform a second retrieval session. NC patients showed worse performances in each session relative to controls. Also, whereas controls showed a linear improvement across all the sessions, NC patients exhibited a slower enhancement, with a significant improvement that occurs only after 7 d from the first retrieval session. Interestingly, further analysis highlighted that patients who improved less across sessions were those who experienced SOREM episodes.

Mazzetti and colleagues [38] investigated the consolidation of procedural motor skills in narcolepsy. Using the same experimental design as Cipolli et al. [62], 14 first-diagnosed NC patients and 14 controls were tested in the FTT. Similarly, to the previous study, NC patients performed worse than controls at all sessions. In addition, both groups showed a steady performance improvement across sessions. However, whereas controls showed a linear performance enhancement across retrievals in line with other studies that used the same task [33–41], NC patients showed a reduced performance change across sessions.

Overall, these two studies highlighted that NC patients experience difficulties in consolidation of both visual discrimination and motor skills. However, no specific association between sleep

features and memory indices was observed in these patients. Therefore, whether these difficulties were the mere expression of difficulties in the consolidation process or reflected a more general memory impairment remains unclear. Indeed, the lack of a wake control condition makes it impossible to attribute these differences to sleep-dependent processes rather than a general consolidation process. Also, although patients showed no impairment in executive functions or general intelligence level, they performed poorly in both procedural tasks before sleeping. Thus, the slower and reduced improvement may depend on this initial lower encoding level. Since these studies did not focus on declarative memory, further studies are required to assess the level of consolidation of verbal and spatial hippocampus-dependent memories. Moreover, given the strong association between narcolepsy and emotional states [65] that may elicit cataplectic responses, it will be clinically useful to investigate whether and how emotional memories, which seem to rely on both REM [66] and NREM sleep [67,68], are consolidated.

Memory consolidation in parasomnias

Parasomnias are sleep disorders characterized by behaviors (e.g., talking, walking) or experiences (e.g., hallucinations, confused states) that occur during sleep or during transitions to and from sleep. NREM parasomnias seem to be caused by "the breakdown of the boundaries between the wakefulness and sleep regulatory systems during slow-wave sleep" [69], whereas REM parasomnias (e.g., RBD, sleep paralysis) are often due to loss of muscle atonia during REM sleep [70].

To date three studies have investigated the offline memory consolidation in parasomnias. Oudiette and colleagues [71] studied whether during sleep, patients with REM sleep behavior disorder (RBD) and sleepwalkers would spontaneously reenact a sequence of movements learned during a pre-sleep training session. Twenty RBD patients ($M_{age} = 66.5$ y), 19 sleepwalking patients ($M_{age} = 34.4$ y), and 18 healthy controls ($M_{age} = 57.9$ y) participated in the study. The protocol included, after an initial adaptation night, an evening learning session of a modified version of a serial reaction time task, followed by video-PSG recording and a retrieval session the next morning. The three groups showed a better performance in the morning relative to the evening, with sleepwalking patients exhibiting a faster performance than controls and RBD patients. However, this effect was probably due to strong age differences between groups. Indeed, as stated by the authors, this study was not designed to directly evaluate memory consolidation in these patients, but to increase the chance to observe task-related behaviors during the post-training sleep. Regarding this aim, one sleepwalking patient did show a behavioral pattern during sleep that resembled the motor sequence trained during wakefulness, suggesting that motor information is indeed replayed during SWS.

In a subsequent study from the same group, Uguccioni and colleagues [72] investigated whether RBD patients could replay part of the sentences learned before sleeping during sleep-talking episodes. Also, they aimed to assess the offline memory consolidation of the verbal material learned before sleeping. A sample of 20 RBD patients ($M_{age} = 67$ y) and 10 controls ($M_{age} = 64.2$ y) was recruited. Participants were tested in two verbal tasks (the free and cued selective reminding, and a modified story recall test) in the evening and during the following morning, after a nocturnal video-PSG. In order to assess daytime consolidation, a subgroup of patients (n = 9) learned a new version of both tasks at the end of the retrieval morning session, which was re-tested later that evening. Then, this subgroup of participants spent a second night in the lab and a retrieval session for the new version of the tasks was performed the following morning. Both RBD and controls showed an offline consolidation effect, improving their performance in both verbal tasks from the evening to the morning session. No differences between groups were observed. Worse verbal performance was observed after daytime consolidation in the subsample of RBD patients, who showed again an overnight consolidation effect when retested the second morning relative to the second evening performance. Regarding sleep-talking episodes, one RBD participant referred to a sentence learned prior to sleeping while in REM sleep. Indeed, this study shows that verbal information encoded prior to sleep may be replayed during REM sleep. Despite most of the evidence indicating that neural replay occurs during NREM sleep (see [73]), neuroimaging studies in healthy individuals have shown that the same cerebral areas active during the execution of a motor task (i.e., the serial reaction time task) in wakefulness were also active during the subsequent REM sleep period [74,75]. Also, a recent study testing pharmacoresistant epileptic patients shows the same stereo-EEG pattern recorded during wakefulness while performing a voluntary movement and during subsequent REM sleep [76]. These results, coupled with animal findings [77,78], suggest that memory replay could also occur during REM sleep.

Using a similar experimental design, the same researchers investigated overnight declarative memory consolidation in 19 patients ($M_{age} = 30$ y) with sleepwalking (n = 16) or sleep terror (n = 3) compared to 19 controls ($M_{age} = 32$ y) [79]. Although the group of patients showed a more fragmented SWS than controls, both groups exhibited an overnight consolidation effect, improving their performance in both the verbal tests used from the evening to the morning session. No significant differences between the patients and controls were observed. Similar to the previous study, verbal performance was comparable between patients and controls after a daytime consolidation period. These results suggest that individuals experiencing parasomnias may not suffer from impairment in memory consolidation.

A comprehensive perspective of memory consolidation in sleep disorders

Overall, the current literature indicates that not all sleep disorders are associated with impairment in memory consolidation. As interestingly proposed by Uguccioni and colleagues [79], memory consolidation impairment may occur when sleep time is actually reduced and/or when sleep architecture is altered (as in insomnia, OSA, and narcolepsy), but not in disorders mainly characterized by alterations of the boundaries between sleep and wakefulness with no marked disruption in sleep organization. Indeed, these studies suggest that the marked sleep fragmentation characterizing insomnia, OSA and narcolepsy may significantly impact hippocampus-dependent memories (i.e., verbal and visual memory), whereby consolidation mainly relies on processes such as memory reactivation, integration, and reorganization (i.e., qualitative changes in memory content; see [80]) are purported to occur during NREM sleep [1,15–17,19,81], but see [82] about the role of REM sleep in memory reorganization]. Similarly, *explicit* MSL, which combines procedural and declarative information and thus is affected to some extent by sleep-related processes [83–85], seems to be impaired in insomnia, OSA, and narcolepsy. On the other hand, individuals with sleep disorders do not show impairments in *implicit* MSL, which seems to not rely on sleep-related processes [86–88] and thus may not be affected by disturbed sleep. Also, results from the MA task, which learning is considered "implicit in nature" [85], are not consistent in the sleep disorders literature. This is in line with studies in healthy samples showing that consolidation of MA abilities is more affected by the passage of time, rather than sleep itself [85,89]. Indeed, it has been suggested that consolidation of explicit motor information may rely on the

interaction between cortico-striatal hubs and the hippocampus during sleep [84], whereas offline consolidation of implicit motor memories mainly relies on the cortico-cerebellar system [85,90], which is unaffected by sleep [91].

These data are in line with Cellini and McDevitt's [32] suggestion that memory retention and forgetting may depend on post-learning level of arousal. Indeed, arousal level during sleep not only is markedly high in insomnia, OSA, and narcolepsy patients, but it also negatively correlates with performance improvement [50,51,56]. Thus, greater arousal levels experienced by these patients may impair their consolidation processes, inducing performance deterioration (i.e., forgetting) or reducing the extent of offline stabilization. Therefore, arousal may affect memory consolidation in several ways. For example, memory reactivation, which is supposed to occur during NREM sleep [1], may be continuously disrupted by arousal bursts and awakenings. Hence, it is plausible that some of the cortical networks may remain weak and unstable, making them a target for the sleep-dependent synaptic depression [9]. Also, it is possible "errors" may occur during each step of this process: memories can be partially reactivated, they can be not completely integrated with existing networks, or memory reorganization (which to date has never been studied in sleep disorders) can be disrupted. In this view, it has been recently shown that in OSA patients the increase of sleep spindles typically induced by pre-sleep learning is not observed [92]. Impairments may also arise from alterations during the downscaling process. For example, it has been observed that insomniacs exhibit excessive glutamatergic activation in response to motor exercise (i.e., thumb movements), which seems to induce increased cortical excitability that can result in a maladaptive homeostatic mechanism. This, in turn, would affect the offline memory consolidation process [93]. Notably, when sleep-disorder patients are successfully treated [54], and the arousal level during sleep decreases, their ability to successfully stabilize information, at least for the motor domain, is restored. This idea of memory impairment due to excessive arousal levels may also account for the less consistent results observed in insomnia. Indeed, while OSA is, by definition, a disorder characterized by high levels of arousal, insomnia is an heterogeneous disorder constituted by several subtypes [23], which are not all characterized by objective alterations of arousal during sleep. Similarly, this idea is also consistent with the absence of memory impairments in patients with parasomnias. These patients are indeed characterized by arousals during the parasomnia episode (e.g., talking, walking), but their sleep architecture does not show any marked alteration (but see [94]), and the main symptomatology they express (e.g., moving during sleep) may just be a manifestation of the mechanisms of memory reactivation (e.g., motor sequence, verbal rehearsal) which becomes "visible" when the boundaries between sleep and wake weaken.

An alternative (and compatible) explanation to the arousal-dependent hypnic alteration is that in patients with insomnia, OSA and, narcolepsy the total sleep and/or specific sleep stage duration may not be sufficient to allow for an optimal consolidation process [58]. Similarly, a disrupted NREM-REM cycle organization may impact memory processes [95,96]. For example, in patients with narcolepsy the occurrence of SOREM, which occurs in about 45% of the sleep episodes [97], is associated with an altered hypnic organization (i.e., longer NREM-REM sleep cycles duration, lower increase in REM duration across sleep cycles; [98]), which may lead to suboptimal memory consolidation processing. This idea is also consistent with a recent model proposed by Vyazovskiy and Delogu [99], which hypothesizes that NREM and REM sleep are complementary stages whose purpose is to "recover" the functionality of neural networks (i.e., synaptic restoration) and "select" which neural network has already terminated the recovery process. Therefore, a disrupted NREM-REM alternation may impair this recovery function, affecting optimal functioning of these cortical networks during the subsequent waking. In this view, Cipolli and colleagues [62] observed greater memory consolidation impairment in patients with narcolepsy with SOREM compared to patients without SOREM.

Effect of treating sleep symptoms on memory consolidation

If memory impairments in individuals with sleep-disorders are the consequence of their sleep symptomatology, it is reasonable to speculate that treating these symptoms may restore, at least partially, memory functioning. Schabus and colleagues [100] tried to test this idea using a neurofeedback training on sensorimotor rhythm (SMR; 12–15 Hz) during wakefulness in 24 primary insomniacs (M_{age} = 34.83 y). After 10 d of SMR conditioning, insomniacs exhibited improved objective and subjective sleep quality. However, either after the SMR conditioning or 5 d of control rhythm, insomniacs showed significant overnight forgetting to a WPA test. Despite this, authors observed a positive association between individuals' SMR-enhancement and the respective overnight memory consolidation. These very interesting and preliminary results suggest that improving the sleep quality in primary insomniacs could also enhance their memory consolidation. Unfortunately, a subsequent study of the same group did not confirm the benefit of this training for both sleep quality and memory consolidation improvement [101], therefore leaving the question of whether improved sleep would ameliorate memory deficits in these patients unanswered. The two studies in OSA patients described above [53,54] highlight two important points. First, cognitive difficulties may be restored to an optimal level when sleep symptomatology is successfully treated [54]. This promising result may have a great impact on sleep medicine, incentivizing the elevated number of potential patients who do not usually seek medical attention [5] to look for treatments in order to address not only sleep symptomology but also cognitive difficulties. Consequently, this may increase general health, well-being, and quality of life of these individuals, while concurrently reducing the direct and indirect societal costs of sleep-related cognitive problems. Second, patients using the CPAP for the first time do not show any overnight memory benefit [53,54], suggesting that the memory restoration process is not immediate, but it requires time. This latter point also prompts an interesting and unexplored question: What are the temporal dynamics of sleep-dependent memory restoration? Although to date no longitudinal studies have described the trajectory and the magnitude of the effect of CPAP (and other) treatments on sleep-related memory consolidation, several studies using standard neuropsychological assessment of cognitive abilities have tested the long-term impact of CPAP on cognitive functions. For example, Zimmerman and colleagues examined the impact of different level of CPAP adherence in patients with OSA who showed verbal memory impairment before the treatment [102]. After three months of CPAP treatments, 68% of the 19 patients with optimal CPAP adherence (individuals who used CPAP an average of >6 h per night) recovered their memory functions, compared to the 44% of the 25 moderate CPAP users (average CPAP use 2–8 h per night), and the 21% of 14 poor CPAP users (average CPAP use < 2 h per night). Of note, the optimal CPAP users showed a probability to recover normal verbal memory 8 times greater than patients with poor CPAP compliance.

On the other hand, in patients with OSA, the long-term effect of treatments on executive functions remains controversial. For example, Antic and colleagues [103] showed that after 3 mo of CPAP treatment, 141 patients with OSA improved their executive functions (i.e., executive maze, verbal recall) compared to baseline,

regardless of the treatment adherence, even if not all participants reached the performance level of healthy controls. In contrast, Saunamäki et al. [104] reported that 6 mo of CPAP treatment (with an average adherence of 6.2 h per night) did not affect the recovery of dysfunctional executive functions (e.g., working memory, verbal fluency) in 40 patients with OSA. Also, a recent meta-analysis on 13 randomized controlled trials showed that after 1–6 mo of CPAP treatment, patients with OSA exhibited a limited improvement in cognitive functioning [105].

Conclusion

Sleep disorders characterized by increased post-learning arousals and disrupted sleep architecture seem to be associated with offline memory consolidation impairments. These impairments, which may arise in childhood, may affect the development and maintenance of cognitive abilities of these individuals, reducing their health and quality of life, increasing the risks of accidents and, consequently, markedly impacting the direct and indirect societal costs of these disorders. Therefore, future studies should be aimed at 1) discerning, using behavioral, pharmacological and, non-invasive brain stimulation intervention, the impact of distinct sleep characteristics on memory consolidation in both patients with sleep disorders and healthy individuals [106]; 2) disentangling the impact of trait-characteristics of patients with sleep disorders (e.g., brain damage in chronic OSA; [44]) from neural mechanisms of offline memory consolidation; and 3) understanding whether and how these memory impairments may be reduced via treatments targeting the sleep symptomatology. These findings are of primary relevance and may hold far-reaching implications not only for the fields of sleep medicine and cognitive neuroscience, but for society as a whole.

Practice points

1) Sleep disorders characterized by elevated arousal level and disrupted sleep architecture such as insomnia, obstructive sleep apnea, and narcolepsy are associated with impairment in the consolidation of declarative and procedural information.
2) Insomnia patients consistently show verbal memory impairments, but results concerning the consolidation of explicit motor sequence and motor adaptation skills are not consistent among studies.
3) Patients with narcolepsy with cataplexy exhibit difficulties in consolidating both explicit motor sequence memories and perceptual abilities.
4) OSA patients systematically exhibit difficulties in consolidating explicit motor sequences, motor adaptation skills, and declarative memories, but they do not show any impairments in encoding new information nor in consolidating implicit motor skills.
5) Children with sleep-disordered breathing have difficulty with the consolidation of verbal but not implicit motor skills.
6) Successful treatment of obstructive sleep apnea symptoms via CPAP is associated with a functional offline memory consolidation process.
7) Patients with parasomnias, such as sleepwalking, night terrors and REM sleep behavior disorder, do not present any memory impairments.

Research agenda

Additional studies are warranted to:

1) Understand the trajectory of the offline memory impairments, from sleep disorder symptomatology onset to the first observable memory impairment.
2) Clarify which tasks are more sensitive to underlying offline memory difficulties as a function of type and severity of sleep disorder, age of the patients, and presence of physical and psychological comorbidities.
3) Evaluate whether specific insomnia phenotypes and subtypes are more susceptible to memory impairments.
4) Assess in narcolepsy the level of consolidation of hippocampus-dependent information such as verbal and spatial memories.
5) Understand how and whether emotional memories, which are critical in eliciting cataplectic attacks, are consolidated in patients with narcolepsy-cataplexy.
6) Understand, using behavioral, pharmacological, and non-invasive brain stimulation interventions, the impact of distinct sleep characteristics in memory consolidation in both patients with sleep disorders and healthy individuals.
7) Understand, using animal, human and computational models, what is the main cause of memory impairment in patients with sleep disorders. A particular effort should address whether and how sleep fragmentation impacts sleep-related memory processing (e.g., neural replay, memory integration and organization, synaptic plasticity and homeostasis).
8) Investigate (using longitudinal designs) whether improving sleep quality will also result in a restoration or improvement of offline memory consolidation in patients with sleep disorders. How will different sleep treatments (e.g., pharmacological, cognitive-behavioral, CPAP) impact memory processing? What is the time course of this memory restoration process? Do these potential benefits generalize to other cognitive and emotional domains?
9) Investigate in patients with sleep disorders the qualitative reorganization of declarative, motor, and emotional memory traces.
10) Disentangle the impact of trait-characteristics of patients with sleep disorders (e.g., brain damage in chronic OSA) from neural mechanisms of memory consolidation during post-training sleep.
11) Future studies should include daytime control conditions in order to understand the extent to which observed memory impairments are specifically related to sleep.

Conflicts of interest

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Appendix A. Supplementary data

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